

|                                | Execution<br>check | $\Delta \text{ loss}$<br>> 0.02 | $\Delta \text{ gradient}$<br>> 0.05 | $\Delta \text{ param update}$<br>> 0.03 |
|--------------------------------|--------------------|---------------------------------|-------------------------------------|-----------------------------------------|
| Execution                      | ✓                  | —                               | —                                   | —                                       |
| Forward-value<br>(numeric)     | —                  | ✓                               | ✓                                   | ✓                                       |
| Gradient<br>(grad_wrong)       | —                  | —                               | ✓                                   | ✓                                       |
| Param. update<br>(param_wrong) | —                  | —                               | —                                   | ✓                                       |